

Currency Premia and Dealer Positions in FX Futures

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1 Purpose

This project constructs a monthly panel linking currency premia, short-rate differentials, and dealer positioning in foreign-exchange futures markets. The measure is intended to summarize how intermediary balance-sheet positions in currency futures covary with exchange-rate movements and expected currency returns.

2 Data

Exchange rates, short-term interest rates, and global controls are obtained from public FRED series, except for BRL, where the short-rate proxy is the Banco Central do Brasil CDI series from SGS. Dealer positions and open interest are obtained from the CFTC Traders in Financial Futures reports. The current sample includes AUD, BRL, CAD, CHF, EUR, GBP, JPY, MXN, NZD, and ZAR against the US dollar. Currencies are included when the exchange rate, the short-rate differential, and the CFTC futures-position series can be aligned at monthly frequency.

3 Variable Construction

Let $E_{i,t}$ denote the end-of-month exchange rate for currency i against the US dollar, expressed consistently so that an increase is a depreciation of the foreign currency against the dollar for currencies quoted as local currency per US dollar, with inverted FRED dollar-per-foreign-currency series where needed. The monthly exchange-rate change is

$$\Delta e_{i,t} = \Delta \log E_{i,t}.$$

The short-rate differential is

$$idiff_{i,t} = i_{i,t}^s - i_{US,t}^{3m}.$$

For most currencies $i_{i,t}^s$ is a three-month money-market rate. For BRL it is a CDI short-rate proxy from Banco Central do Brasil SGS series 4389, reflecting the absence of a comparable public three-month BRL series in the FRED/OECD source used for the rest of the panel. Currency excess returns are approximated by

$$rx_{i,t} = idiff_{i,t} - \Delta e_{i,t}.$$

Dealer positions are measured as dealer long positions minus dealer short positions in CFTC Traders in Financial Futures data. The same construction is also reported for asset managers, leveraged funds, other reportables, and non-reportables. The normalized dealer-position measure is

$$f_{i,t}^* = 100 \times \frac{DealerLong_{i,t} - DealerShort_{i,t}}{\overline{OI}_{i,t}^{12m}},$$

where $\overline{OI}_{i,t}^{12m}$ is the 12-month moving average of open interest. The main dealer-position shock is $\Delta f_{i,t}^*$.

4 Empirical Specifications

The script reports three main specifications. First, it estimates cross-sectional relationships between average rate differentials and average dealer positions. Second, it estimates predictive regressions of next-month currency excess returns on dealer positions and rate differentials with currency fixed effects. Third, it estimates concurrent exchange-rate regressions of $\Delta e_{i,t}$ on $\Delta f_{i,t}^*$, changes in interest differentials, and global controls. Standard errors use HAC covariance estimates. As supplementary survey-expectations checks, the project collects ECB Survey of Professional Forecasters assumptions for USD/EUR and Banco Central do Brasil Focus expectations for BRL/USD. These public survey series do not reproduce the Consensus Economics three-month expectations used in IMF WP/25/153, but they allow the project to distinguish observed currency premia from survey-implied expected exchange-rate changes for two currency pairs. The project also reports a market-implied approximation based on continuous front CME FX futures. This futures basis is defined as the log difference between the front-futures price, converted into foreign currency per US dollar, and the spot exchange rate. Because the futures contract is a rolling front contract, the measure should be interpreted as a front-futures basis rather than a constant-maturity forward premium. As a robustness check, the script also subtracts the short-rate differential multiplied by an assumed time to maturity. Two assumptions are reported: maturity to the next quarterly IMM expiry and a fixed three-month maturity. The figures emphasize the cross-currency median, interquartile range, and advanced-economy versus emerging-market medians. This produces a transparent futures-implied FX basis proxy, but not a contract-level CIP deviation. To assess the quality of the futures premium leg, the project compares the annualized futures premium with the three-month market-implied forward premium, $\rho_{i,3m,t}$, published by Du, Keerati, and Schreger. The comparison uses their forward-premium variable only, not their government-bond CIP deviation, because the rate leg in this project differs from the government-bond yield differential in their dataset. A validation grid evaluates day-count conventions, expiry assumptions, maturity windows, and volume thresholds against the DKS forward-premium variable. The preferred advanced-economy rule uses observations where the front futures contract has 60–80 days to assumed IMM expiry, ACT/360 annualization, and monthly median futures volume of at least 10,000 contracts. This liquidity and maturity screen is transparent and does not use the DKS values to mechanically calibrate the proxy; it improves the advanced-currency comparison but leaves too little reliable evidence for emerging-market futures.

5 Outputs

The replication script writes a monthly panel, coverage diagnostics, per-currency estimates, panel regression summaries, and charts. The figures use a consistent project palette and are regenerated by the monthly GitHub Action.